Communities and Crime Analysis Dashboard

Project Overview and Objective:

This project focuses on cleaning, transforming, analysing, and visualizing Communities and Crime data using Excel, Power Query, and Power BI.

The dataset contains population, demographic, socioeconomic, housing, police, and crime-related attributes for different communities. The project aims to transform the raw dataset into a structured and analysis-ready format and develop an interactive Power BI dashboard.

Main Objectives

- To clean and preprocess raw community and crime data using Excel and Power Query.
- To identify and handle missing values, duplicates, inconsistent data types, and formatting issues.
- To analyse demographic, socioeconomic, housing, police, and crime-related factors.
- To compare crime levels across different communities and states.
- To identify factors that are associated with higher or lower crime levels.
- To create an interactive Power BI dashboard for data-driven analysis and decision-making.

Data Sources:

Source Description and Timeline

Dataset: Communities and Crime Unnormalized Dataset

Source: UCI Machine Learning Repository

Dataset Size: Approximately 2,215 communities and 147 attributes

Data Period: Primarily based on 1990 U.S. Census data and 1995 FBI crime data.

Geographical Coverage: Communities across the United States.

Domain

Crime Analytics / Social Analytics / Public Safety

Problem Statement:

Communities have different population, demographic, social, economic, housing, law-enforcement, and crime-related characteristics, making it difficult to identify meaningful crime patterns from raw data.

This project aims to clean and transform the Communities and Crime dataset using Excel and Power Query, analyse relationships between community characteristics and crime levels, and develop an interactive Power BI dashboard to compare communities and identify factors associated with crime.

Key Analysis Questions

1. Which states or communities have higher crime levels?
2. What demographic and socioeconomic characteristics are associated with crime?
3. How do poverty and unemployment relate to crime levels?
4. How do population density and urbanization differ across communities?
5. How do police-related factors vary between communities?
6. Which communities have higher violent-crime rates?
7. What patterns can be identified between socioeconomic conditions and crime?

Attribute (Column / Features) Details

Attribute Name	Data Type	Description
communityname	Text	Name of the community
State	Text	State in which the community is located
countyCode	Integer	County identification code
communityCode	Integer	Community identification code

Attribute Name	Data Type	Description
fold	Integer	Data fold used for validation
pop	Numeric	Population of the community
perHoush	Decimal	Average number of persons per household
pctBlack	Decimal	Percentage of population that is Black
pctWhite	Decimal	Percentage of population that is White
pctAsian	Decimal	Percentage of population that is Asian
pctHisp	Decimal	Percentage of population that is Hispanic
pct12-21	Decimal	Percentage of population aged 12–21
pct12-29	Decimal	Percentage of population aged 12–29
pct16-24	Decimal	Percentage of population aged 16–24
pct65up	Decimal	Percentage of population aged 65 and above
persUrban	Numeric	Number of people living in urban areas
pctUrban	Decimal	Percentage of population living in urban areas
medIncome	Numeric	Median household income
pctWwage	Decimal	Percentage of households receiving wage/salary income
pctWfarm	Decimal	Percentage of households receiving farm/self-employment income
pctPoverty	Decimal	Percentage of population below the poverty level
pctUnemploy	Decimal	Percentage of unemployed population
pctCollGrad	Decimal	Percentage of population who are college graduates
popDensity	Decimal	Population density
murders	Numeric	Number of murders
murdPerPop	Decimal	Murder rate per population
rapes	Numeric	Number of rapes
rapesPerPop	Decimal	Rape rate per population
robberies	Numeric	Number of robberies
robbbPerPop	Decimal	Robbery rate per population
assaults	Numeric	Number of assaults
assaultPerPop	Decimal	Assault rate per population
burglaries	Numeric	Number of burglaries
burglPerPop	Decimal	Burglary rate per population
larcenies	Numeric	Number of larcenies
larcPerPop	Decimal	Larceny rate per population
autoTheft	Numeric	Number of auto thefts
autoTheftPerPop	Decimal	Auto-theft rate per population
arsons	Numeric	Number of arsons
arsonsPerPop	Decimal	Arson rate per population
violentPerPop	Decimal	Violent crime rate per population
nonViolPerPop	Decimal	Non-violent crime rate per population

Tools & Technologies

Excel

Used for:

- Initial data inspection
- Data organization

- Filtering and sorting
- Duplicate checking
- Basic data validation
- Pivot Tables
- Initial exploratory analysis

Power Query

Used for:

- Data cleaning
- Handling missing values
- Changing data types
- Renaming columns
- Removing unnecessary columns
- Standardizing data
- Creating transformation steps
- Preparing the dataset for Power BI

Power BI

Used for:

- Data modelling
- Relationships between tables
- DAX calculations
- Interactive visualizations
- Slicers and filters
- Drill-down analysis
- Dashboard creation
- Crime and community analysis

DAX

Used for calculating important metrics such as:

- Total Population
- Average Population
- Total Murders
- Total Robberies
- Total Assaults
- Average Violent Crime Rate
- Average Non-Violent Crime Rate
- Crime Rate comparisons

- State/community rankings

Data Pre-Processing (Excel / Power Query)

Step 1: Fix headers

The original file had no usable column headers. The first row of real column names was promoted to be the table header so every column had a readable, correct name (e.g. larcPerPop).

Step 2: Remove columns with very high missing data

22 columns were removed because they were missing in approximately 84.47% of rows. Columns missing that much data cannot be reliably imputed or analyzed, so they were dropped rather than filled in.

Step 3: Remove non-predictive identifier columns

countyCode and communityCode were removed. Both were about 55% missing, and even where present they are just ID numbers (not characteristics of a community), so they add no analytical value and were safe to drop.

Columns remaining after Steps 2-3: 123 (down from 147).

Step 4: Impute the two main target columns

ViolentCrimesPerPop and nonViolPerPop (the two key crime-rate columns) had missing values filled in using median imputation - each blank was replaced with the median value of that column. Median was used instead of average because crime rates are skewed by a small number of very high-crime communities, and the median is not distorted by those outliers.

Step 5: Check for duplicate rows

The full table was checked for exact duplicate rows, and also for duplicate combinations of communityname and State (the natural identifier for a community). Result: 0 duplicates found either way, so no rows were removed.

Step 6: Check and correct data types

Every column was reviewed and assigned the correct type:

- 37 columns were set to whole numbers (counts, e.g. murders, pop, medIncome)
- 84 columns were set to decimal numbers (rates and percentages, e.g. pctBlack, murdPerPop)
- communityname and State were set to text, with State standardized to uppercase 2-letter codes

Step 7: Handle remaining missing values

After Step 4, 15 columns still had gaps - mainly rarer crime types such as rapes, arsons, and their per-population rates, where some small communities simply had zero reported incidents on record. These were filled using the same median imputation method already used in Step 4, keeping the approach consistent across the whole dataset. After this step, 0 missing values remained anywhere in the dataset.

Step 8: Standardize and clean

- Trimmed extra spaces from community names
- Converted all State codes to uppercase
- Rounded percentage columns to 4 decimal places for consistency
- Confirmed all State codes are valid 2-letter US codes (48 states/DC present in this dataset)

Step 9: Filter and sort

Rows were checked for any blank community name or state (none were found, so no rows needed to be removed). The table was then sorted alphabetically by State, then by community name, to make it easier to browse and to give a predictable, repeatable order.

Step 10: Create calculated fields

Several new columns were added to make analysis easier:

New column	How it's calculated
TotalViolentCrimeCount	murders + rapes + robberies + assaults
TotalPropertyCrimeCount	burglaries + larcenies + autoTheft + arsons

New column	How it's calculated
TotalCrimeCount	TotalViolentCrimeCount + TotalPropertyCrimeCount
PctMinorityPop	pctBlack + pctAsian + pctHisp
CrimeSeverityIndex	(ViolentCrimesPerPop x 2) + nonViolPerPop, weighting violent crime higher
PopulationSizeCategory	pop split into Small / Medium / Large / Very Large using quartiles

Step 11: Build Fact and Dimension tables

To make the dataset Power BI-friendly, it was split into a simple star schema, all linked by a new CommunityKey column:

- Dim_Community - location and core demographic/economic fields (population, income, poverty, land area, etc.)
- Fact_Crime - all crime counts, crime rates, and the new calculated crime fields
- Dim_Socioeconomic - the remaining detailed demographic, housing, and economic columns

Step 12: Validate the final dataset

Before delivering the final file, seven checks were run to confirm the cleaning worked correctly:

Check	Result
Row count still matches original (2,215)	Passed
No missing values remain	Passed
No duplicate CommunityKey values	Passed
No duplicate communityname + State pairs	Passed
All percentage columns fall between 0 and 100	Passed
All State codes are 2 letters	Passed
Fact table row count matches Dimension table row count	Passed

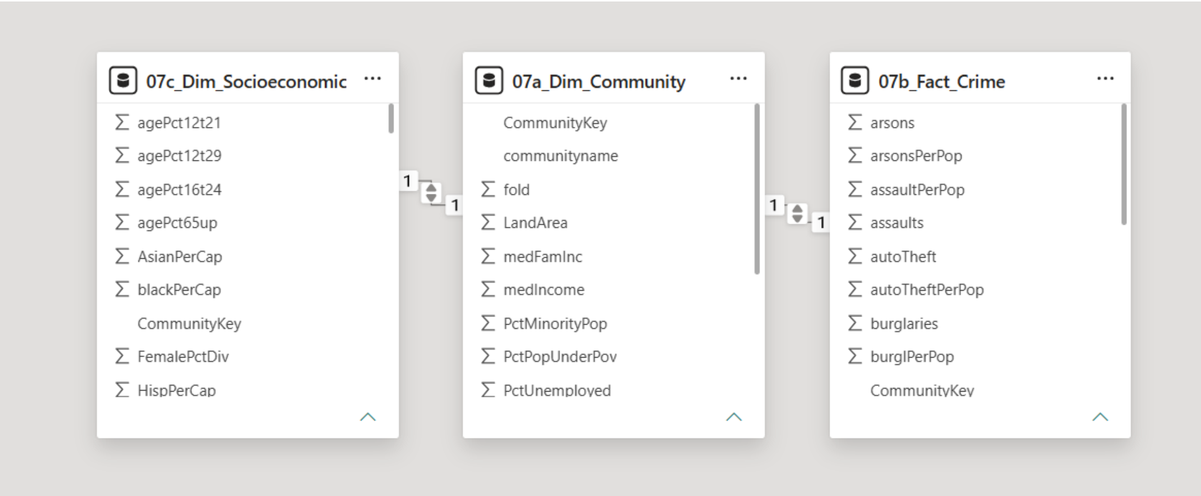
Step 13: Save each stage separately

Every major stage was saved as its own sheet in the workbook, rather than overwriting previous work. This keeps a full audit trail and means no earlier work is ever lost:

Sheet name	Contents
Raw_Data	Original, completely unedited file (preserved as-is)
Headers	Raw data with headers fixed only
Handle Missing Values	Working copy after Steps 2-4 (column removal + first imputation)
01_Duplicate_Check	Duplicate check results
02_Data_Types	Record of which columns were set to which data type
03_Missing_Values_Handled	Log of every column imputed and the median value used
04_Standardized	Notes on cleanup actions taken
05_Filtered_Sorted	Filter/sort summary
06_Calculated_Fields	Documentation of every new formula/column
07a / 07b / 07c	Dim_Community, Fact_Crime, Dim_Socioeconomic tables
08_Validation	Results of the 7 validation checks
09_Final_Cleaned_FlatTable	The final, single flat table - ready for analysis

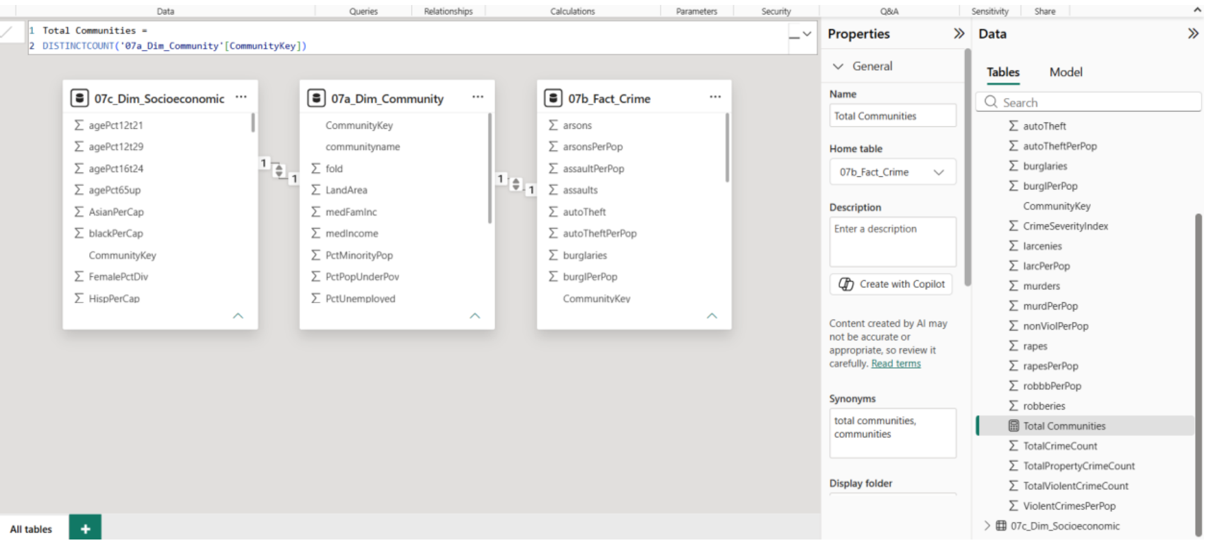
7. Data Modelling and DAX (Power BI)

Relationships

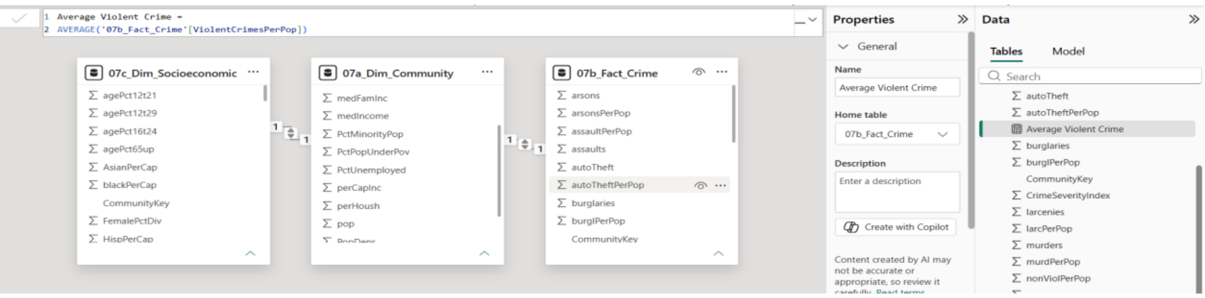


Create DAX Measures

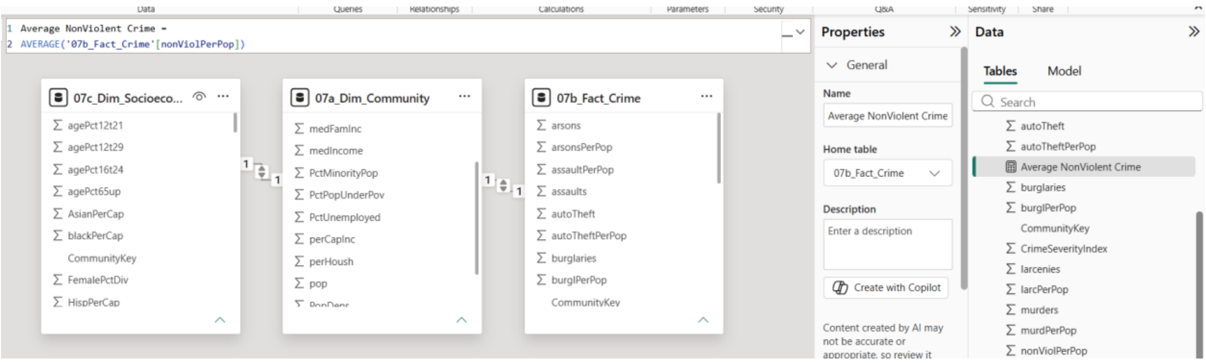
- **Measure 1 — Total Communities**



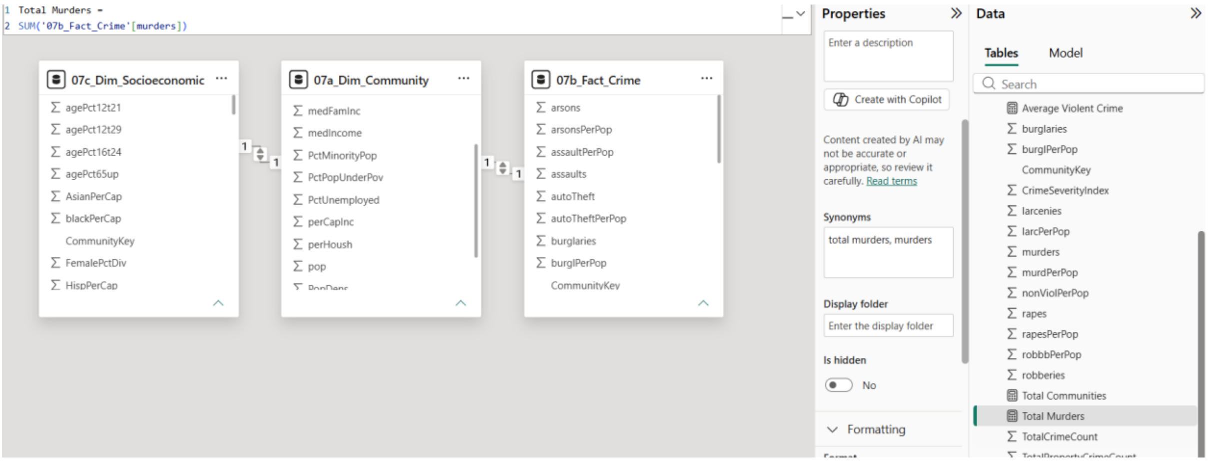
- **Measure 2 — Average Violent Crime**



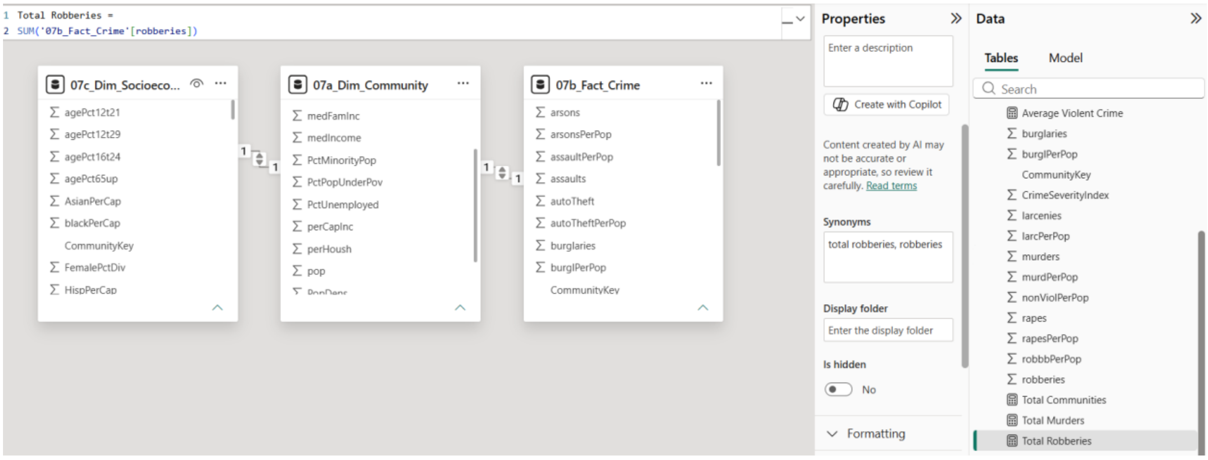
- **Measure 3 — Average Non-Violent Crime**



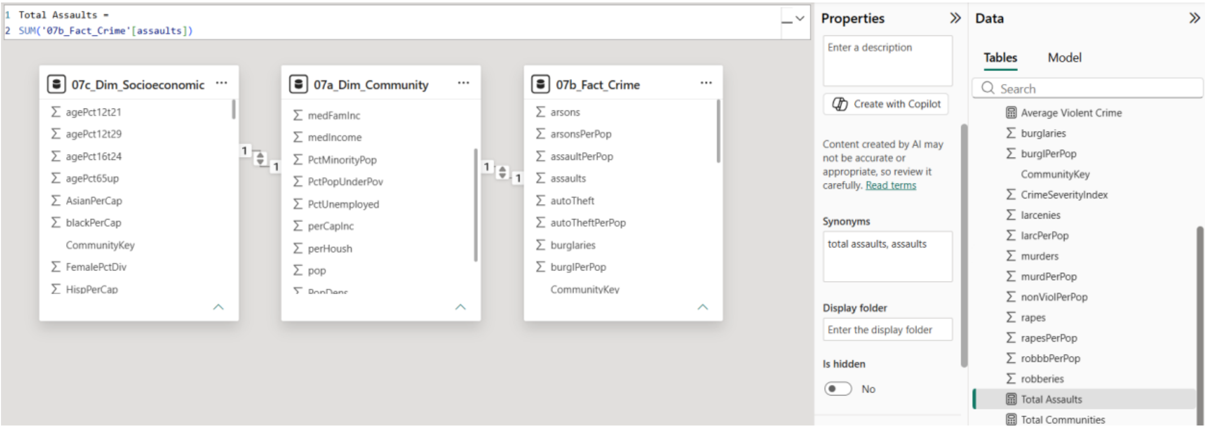
- **Measure 4 — Total Murders**



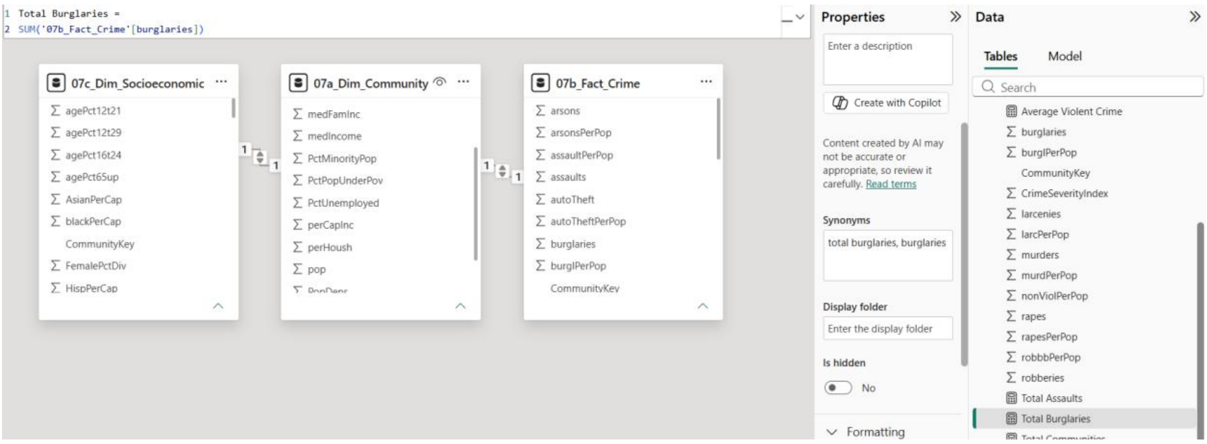
- **Measure 5 — Total Robberies**



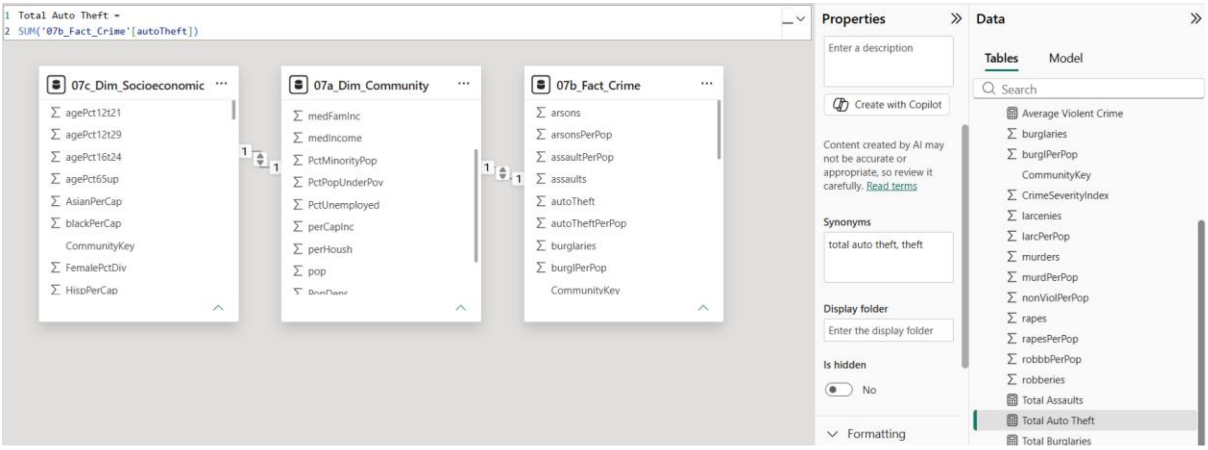
- **Measure 6 — Total Assaults**



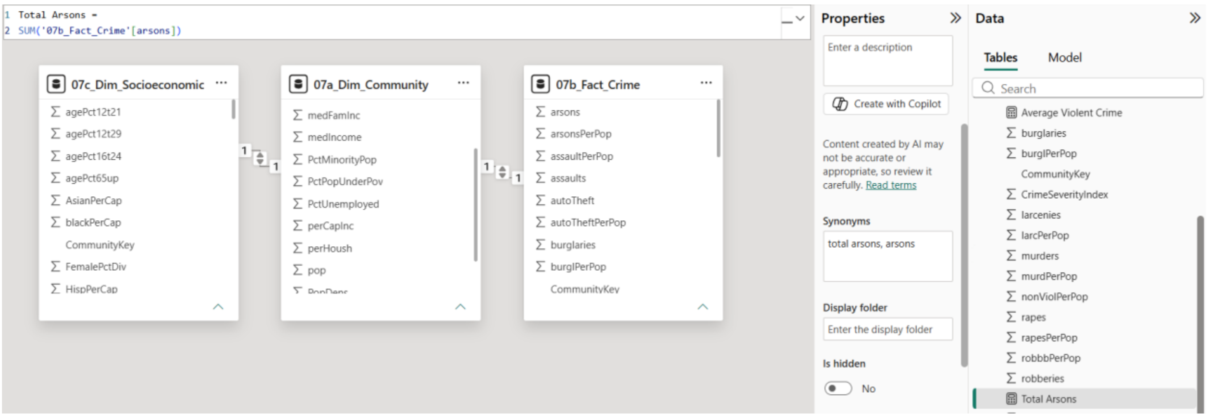
- **Measure 7 — Total Burglaries**



- **Measure 8 — Total Auto Theft**

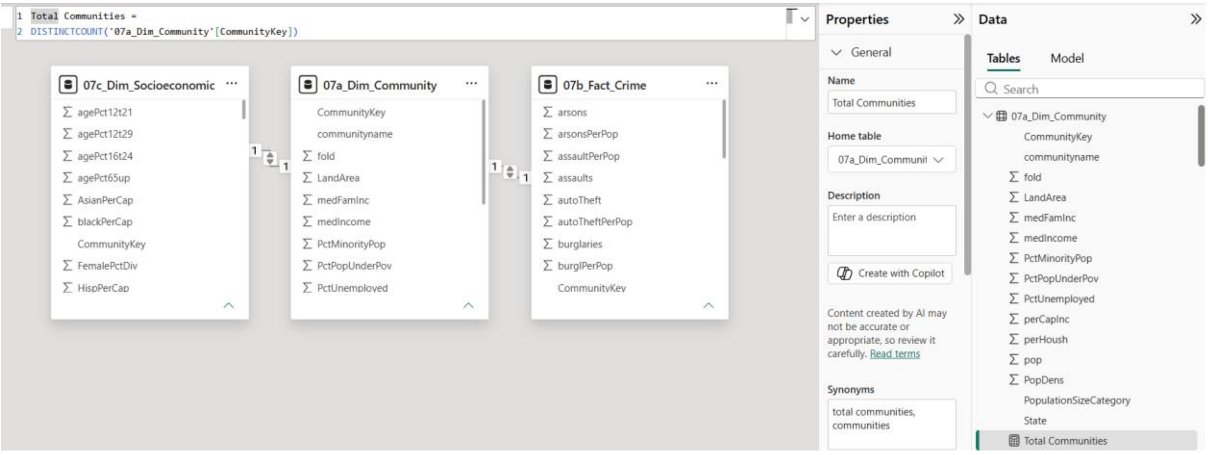


- **Measure 9 — Total Arsons**

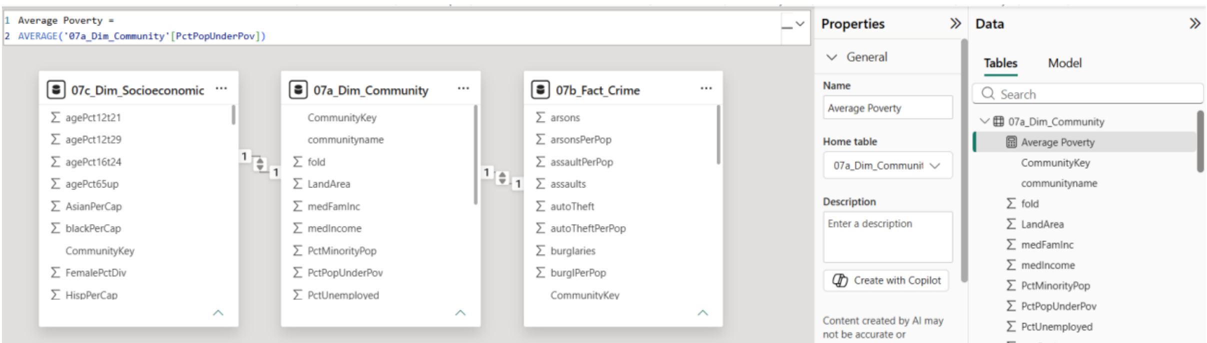


Create Community Measures

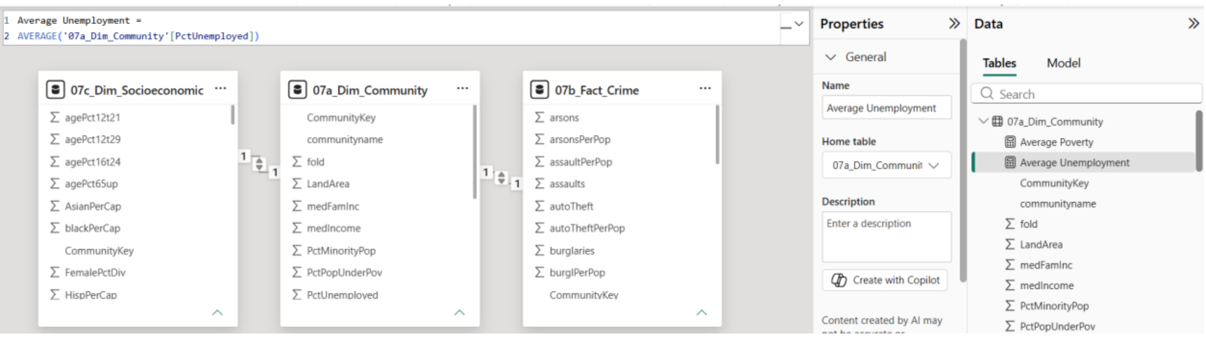
- **Total Communities**



- **Average Poverty**

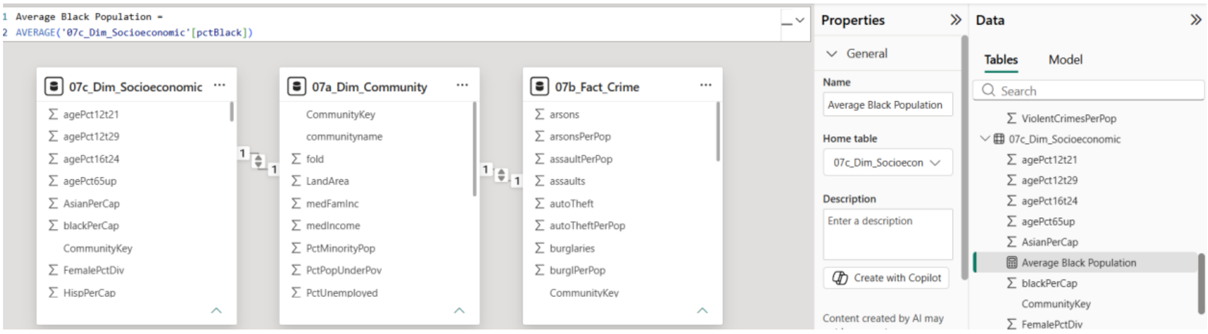


- **Average Unemployment**

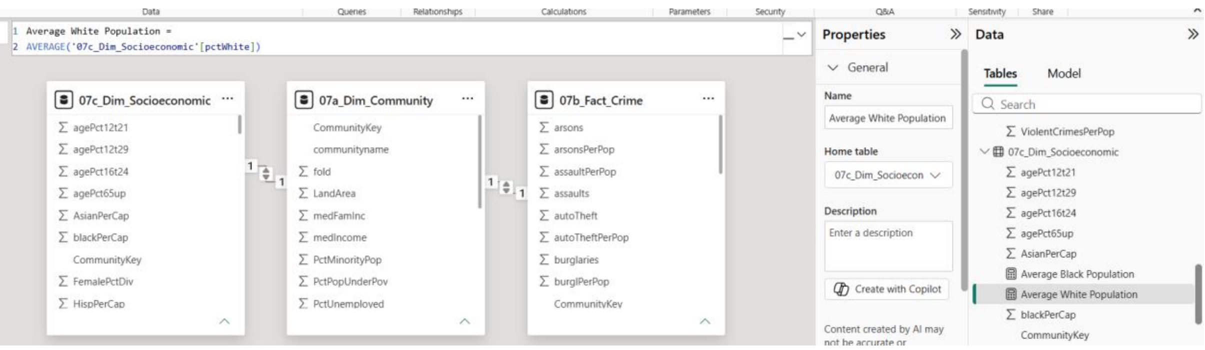


<u>Create Socioeconomic Measures</u>

- **Average Black Population %**



- **Average White Population**



8.Analysis and Visualizations (Power BI)

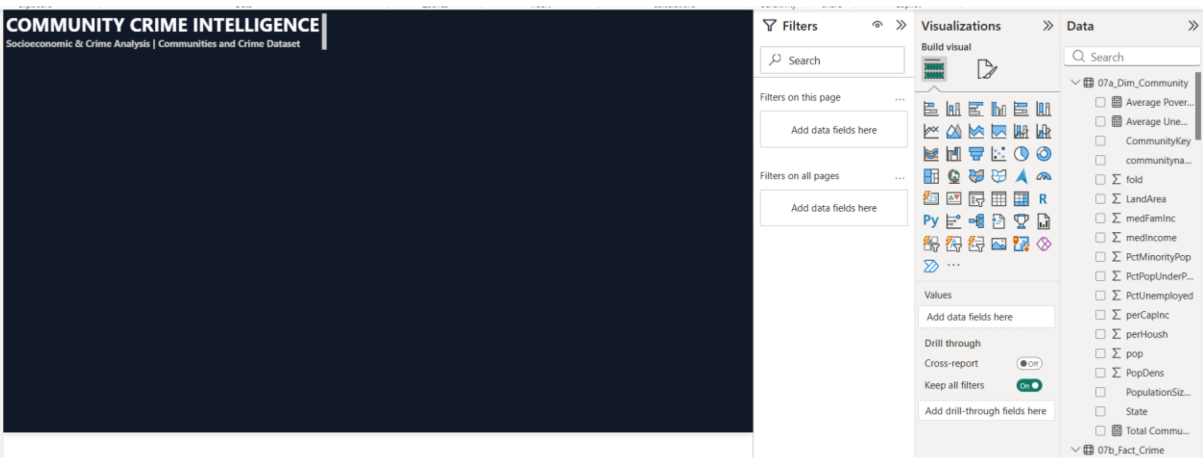
Step 1 — Set the dashboard background

Go to Report View.

Then:

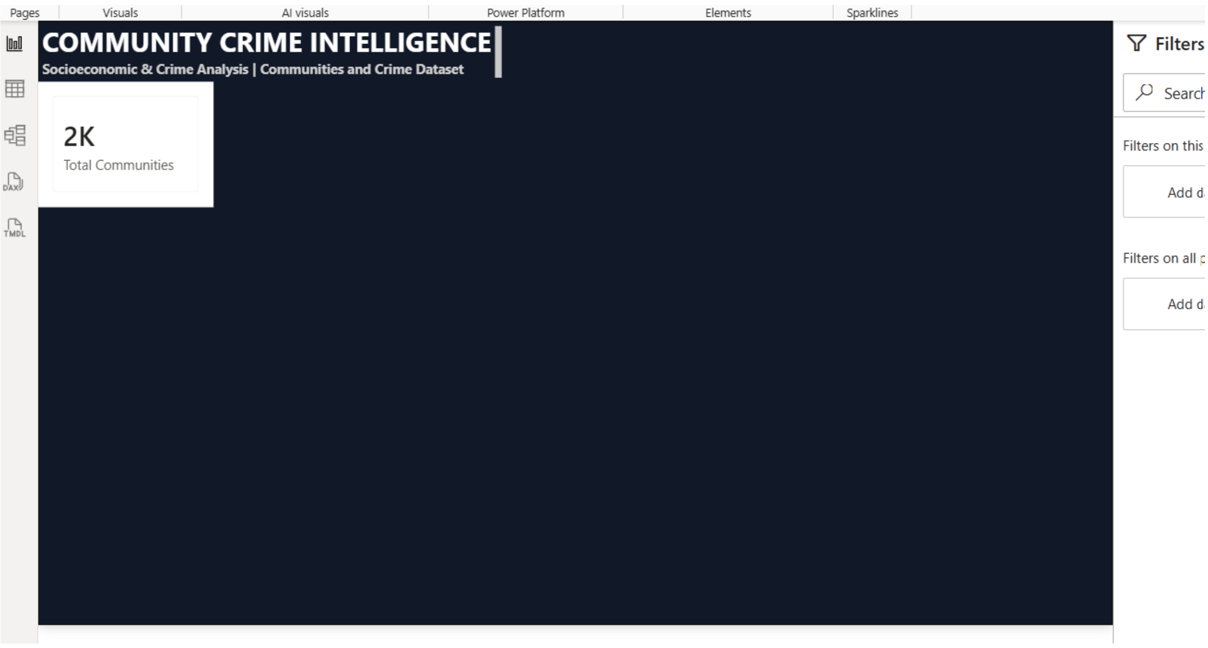
Format → Canvas settings / Page → Background

Set a dark navy/charcoal background.

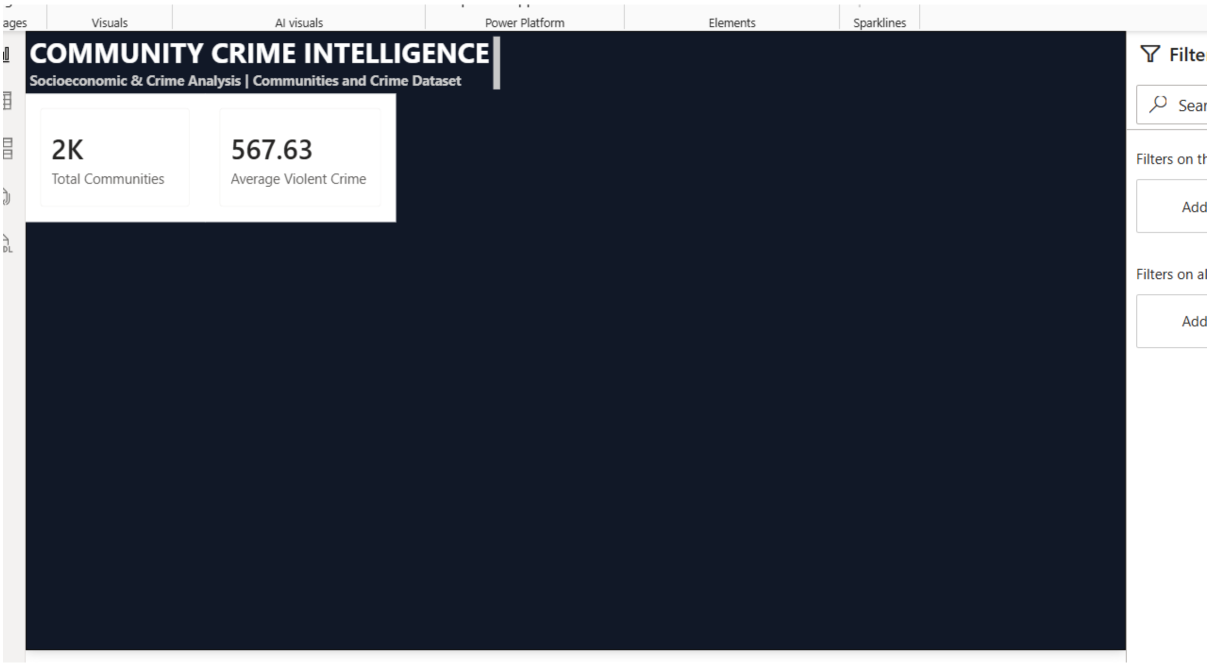


Step 2 — First KPI Card

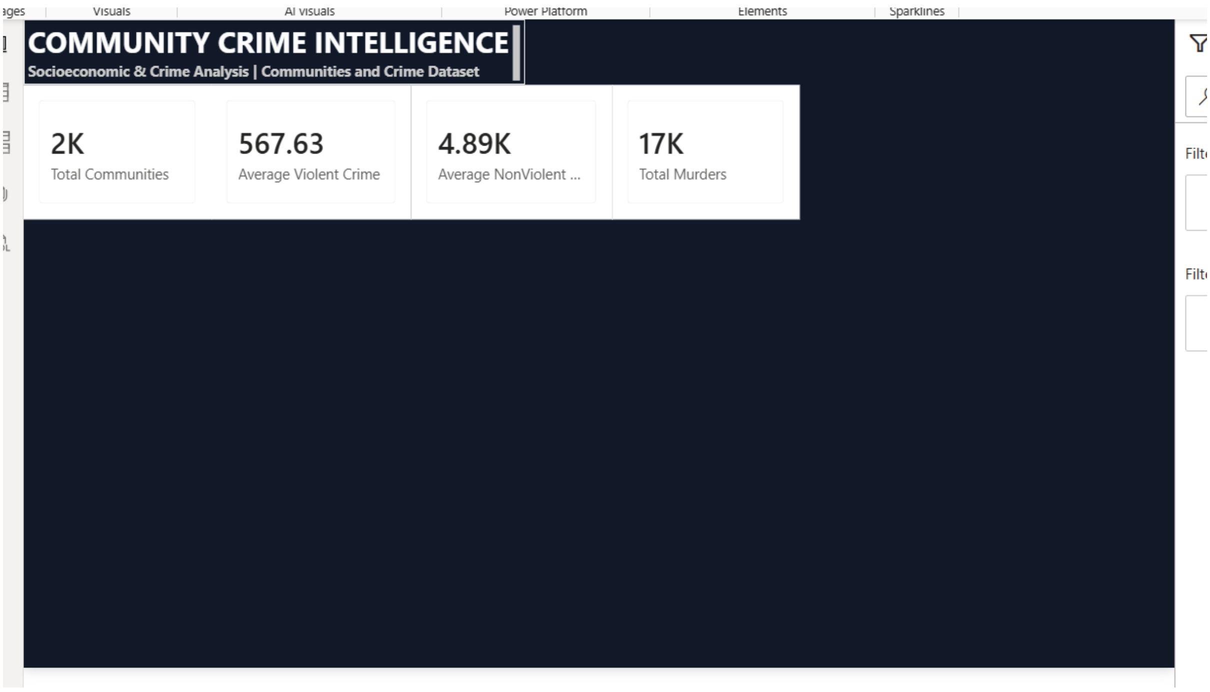
Total Communities



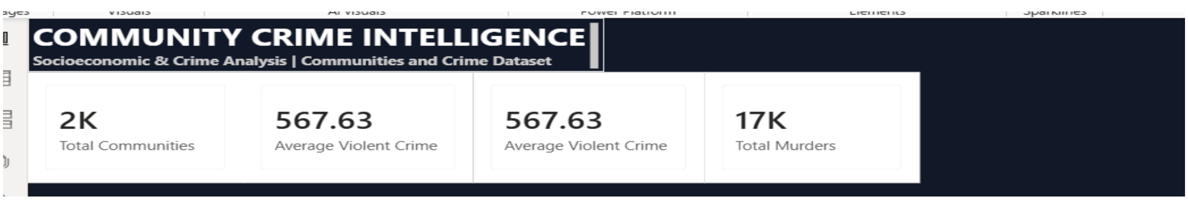
Average Violent Crime



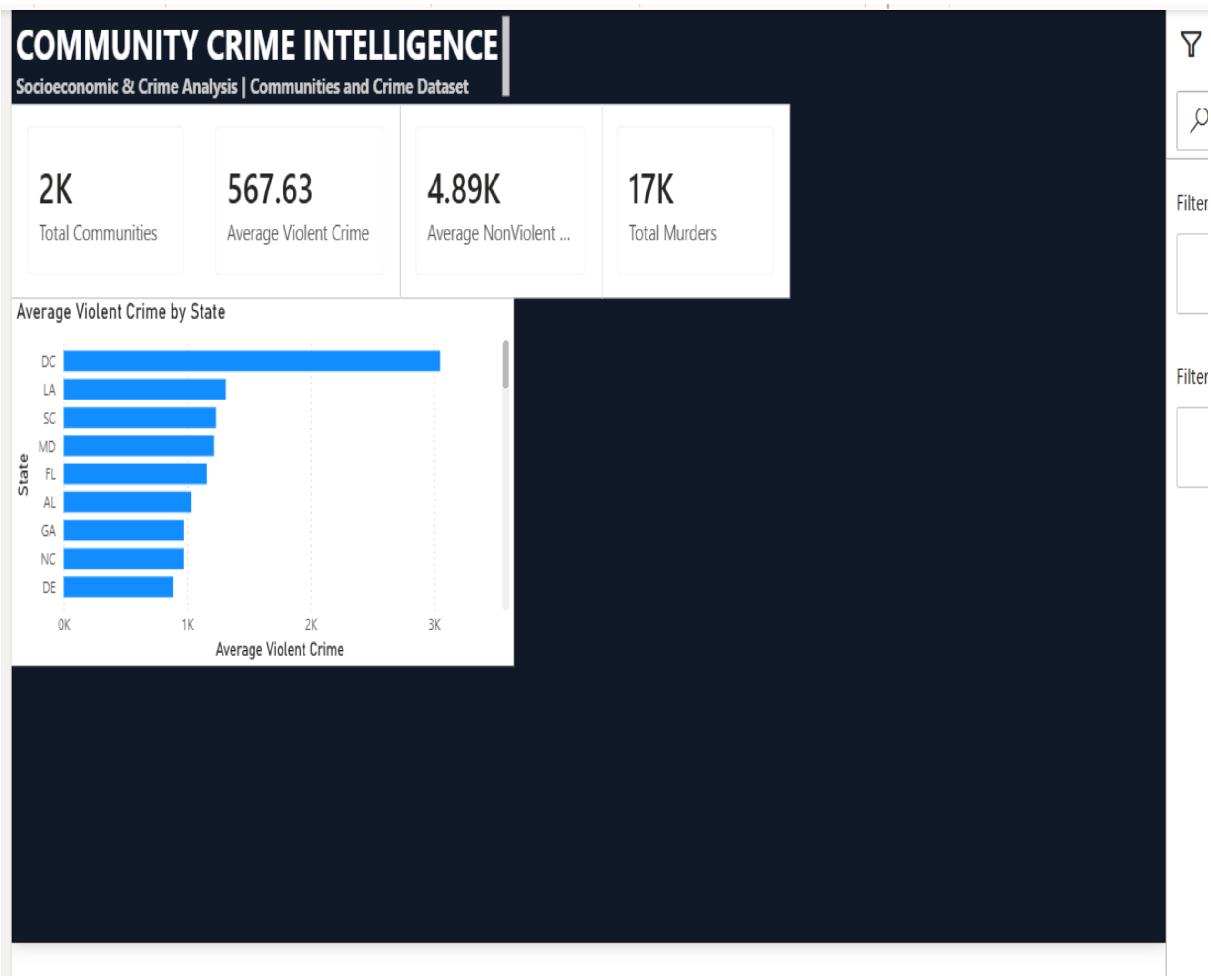
Average Non-Violent Crime



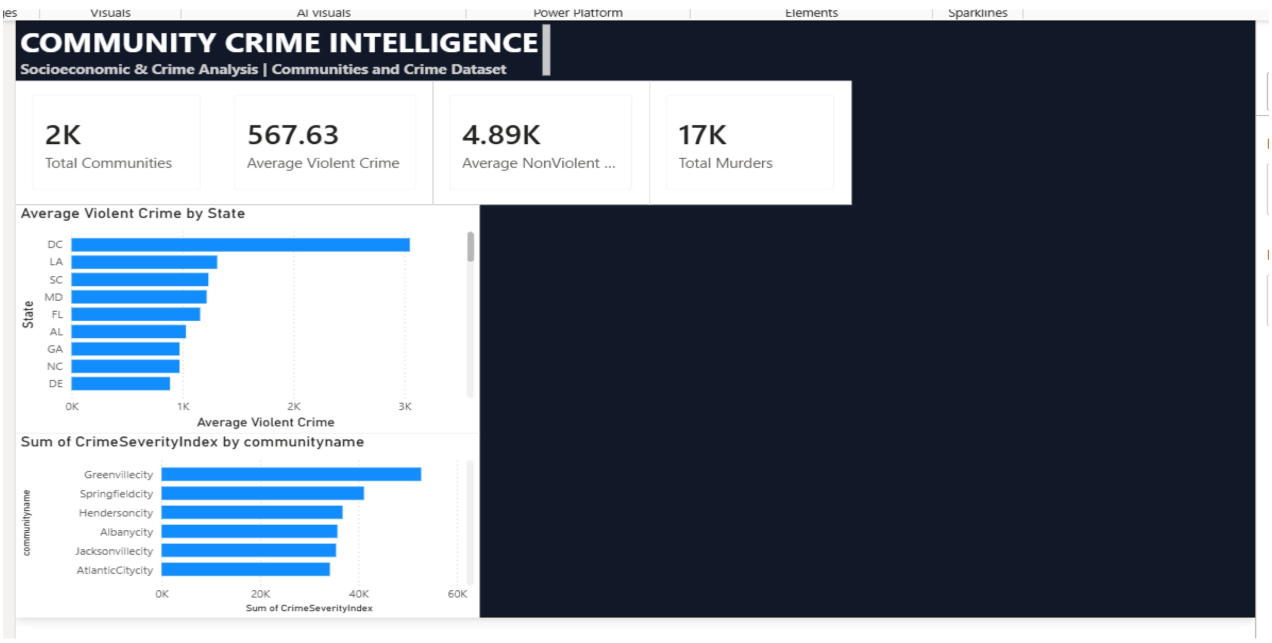
Total Murders



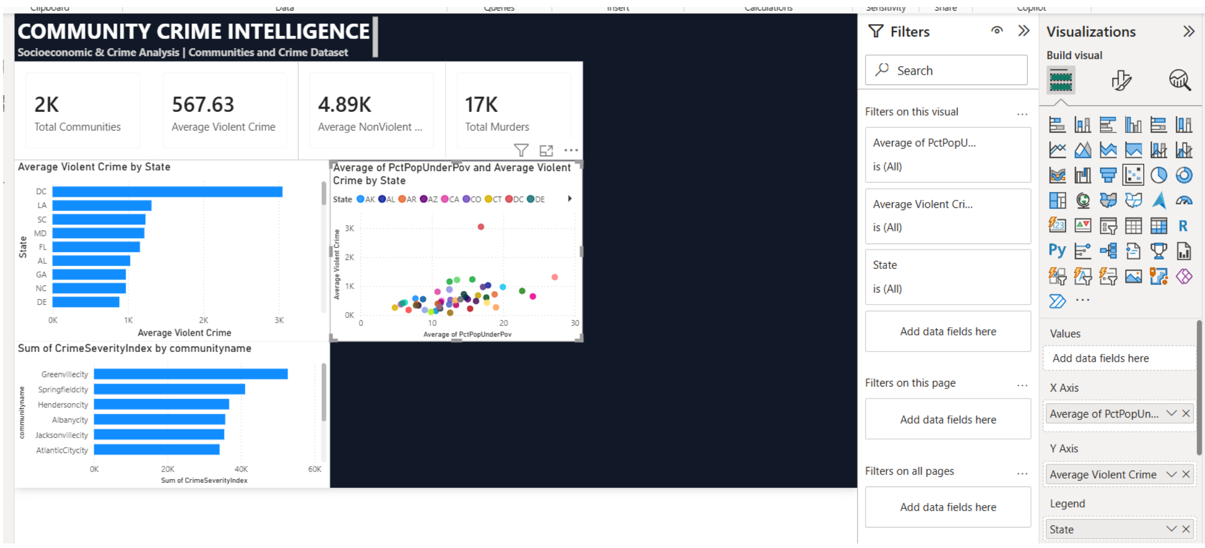
Crime by State



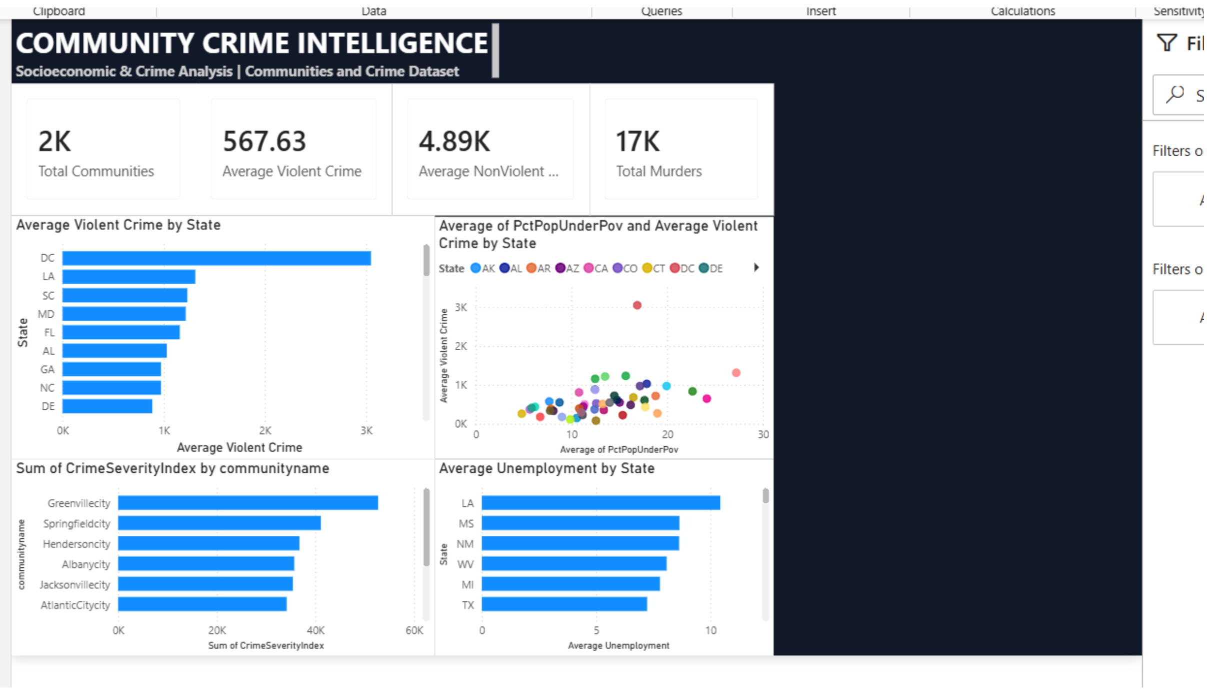
Crime Severity by Community



Poverty vs Violent Crime

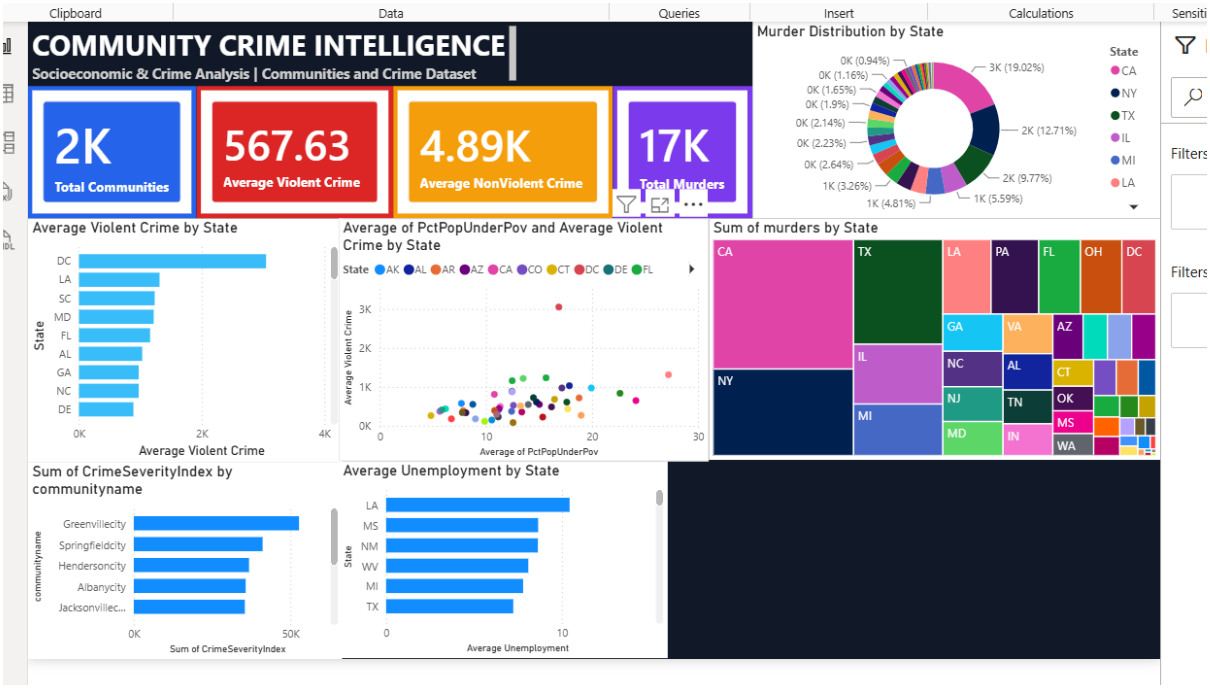


Average Unemployment by State

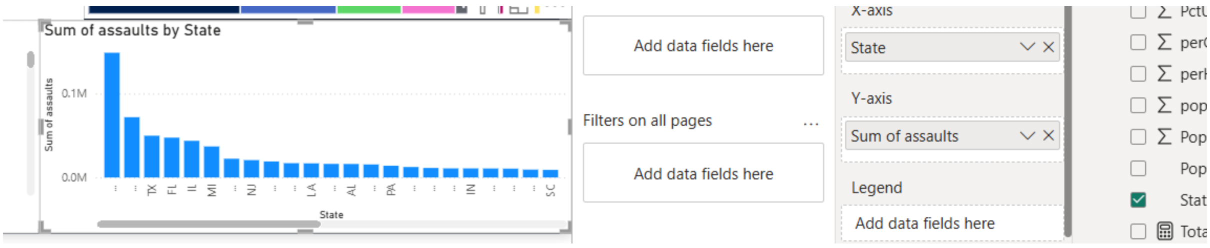


Total Murders by State

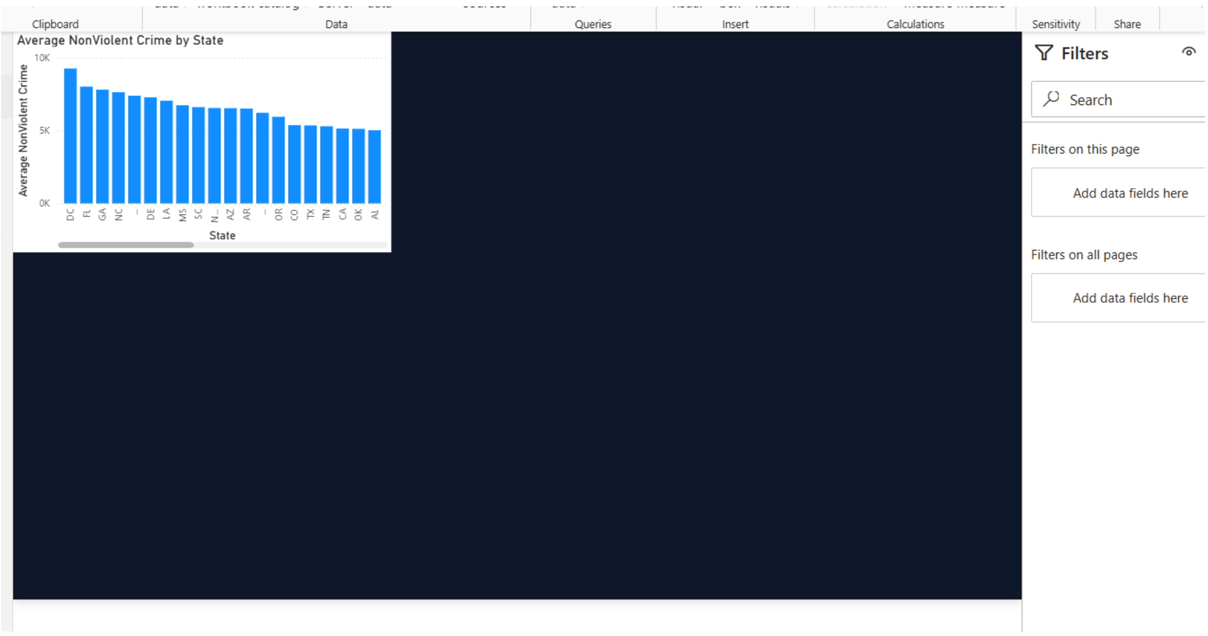
Crime by Type



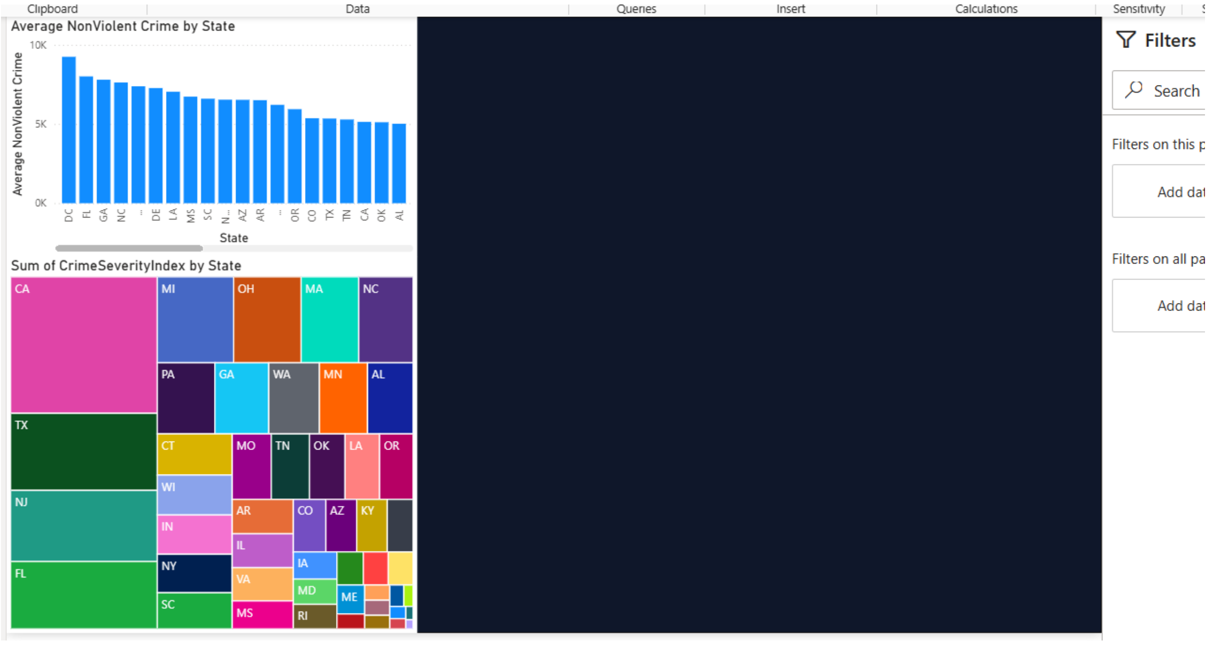
Total Assaults by State



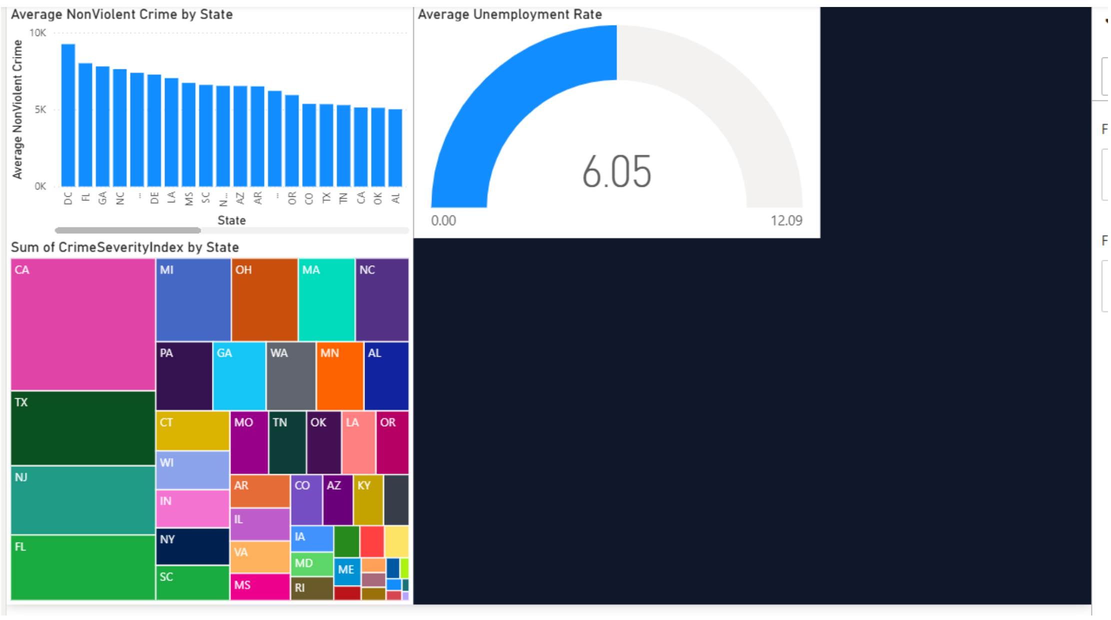
Average NonViolent Crime by State



Crime Severity by State



Average Unemployment



9. Insights & Conclusions

The analysis of the Communities and Crime dataset identified differences in crime levels across states and communities. The dashboard helped analyze the relationship between crime and socioeconomic factors such as poverty and unemployment.

- Descriptive: Crime levels, murders, assaults, poverty, and unemployment were summarized using Power BI visuals.
- Diagnostic: Differences in crime levels were compared with socioeconomic factors to understand possible relationships.
- Predictive: High-risk communities can potentially be identified using crime severity, poverty, unemployment, and population factors.
- Prescriptive: Crime-prevention resources and support can be prioritized for communities showing higher crime severity and socioeconomic challenges.

10. Conclusion

The integration of **Excel and Power BI** provided an effective end-to-end approach for cleaning, modelling, analysing, and visualizing the Communities and Crime dataset. The final dashboard presents key crime and socioeconomic indicators clearly and supports **data-driven decision-making and better identification of high-risk communities**.